FLY BY SNAPSHOTS

Nothing exemplifies the robust nature of India's drone making ecosystem better than these impressive machines



SPOT V3

CDSPACE ROBOTICS, BENGALURU

PAYLOAD: *25 KG*

- ***Runway independent operation with vertical take-off and landing***
- ***Border patrol and surveillance***
- ***Gunfire direction***
- ***ISR for strategic facilities***
- ***Deep sea and coastal reconnaissance***



VAANVILI

GARUDAN, COIMBATORE

PAYLOAD: OVER *4 KG*
MAX DISTANCE: *5 KM*
ENDURANCE: *45 MINUTES*

- ***Unmanned Aerial Vehicle ideal for surveillance, logistics and mapping***
- ***Fully autonomous hexacopter for take-off and landing***

in 2022 to $30.86 billion by 2029. It is estimated that around 2,000 attack drones and over 80,000 surveillance drones will be sold around the world in the next 10 years.

With the rapid evolution of drone warfare, the Indian military is procuring drones in large numbers. The recently concluded Defexpo in Gandhinagar, Gujarat, where over 200 Indian drone manufacturers showcased their products, witnessed a significant interest in unmanned platforms by the military. The military top brass had foreseen the shape of things to come. A call to Indian drone makers was given in 2018, when the IAF launched the Mehar Baba swarm drone competition, which lasted till 2021 and which led to the building of an ecosystem in drone manufacturing. Over 150 Indian start-ups competed for the top honours.

The Indian military and its counterparts worldwide are keenly watching the Ukraine conflict, where drones were used to devastating effect, destroying Russian armour, missile launchers and supply vehicles, and helping blunt the offensive towards Kyiv in the early days of the war. Russia's use of swarms of single-use 'kamikaze' drones in recent days is no less noteworthy. Military analysts say that Azerbaijan got the upper hand in its recent skirmish with Armenia thanks to drones. Currently, Indian forces are dependent on foreign-made drones, like the Israeli Heron and Searchers for ISR purposes. "The Indian army is satisfied with our surveillance drones. Besides India, our focus is towards exports," says 36-year-old Nanda Kumar.

Along with three engineering classmates of BITS Pilani, 31-year-old Satyabrata Sayapathy founded Bon V Technology and they started working on drones in 2021. After 18 months, they developed Bon V, India's first AI-powered multi-utility logistic drone. "Dropping ration to soldiers deployed in high altitude terrain is challenging. The use of helicopters is risky. Our unique logistic drone for high-altitude areas can be used at mountain deployments in the east for China and in the northwest on the LoC by the army," says Sayapathi, who is confident of bagging the army's latest order to procure 350 high and medium-altitude drones.

The army is currently using ideaForge's SWITCH 1.0 at the LAC to keep an eye on Chinese forces

Gaurav Achha, co-founder of Bon V, claims that they have demonstrated the drone's capability before the army in Rupa Valley in Arunachal Pradesh, close to the Indo-China border. "Our drone has done over 50 hours of flying during field trials," Achha says. Bon V's autonomous sensors and data processing powers also allow it to be the ideal craft for combat search and rescue or for evacuation during a disaster. Given the backdrop of the huge military build-up in Ladakh after the 2020 border standoff with China, the Bon V drone does seem to be a vital tool.

Logistics and surveillance aside, the army requires UAVs to add to its sword arm. Alina Usmani, project head of India's first indigenously designed combat drone, Jatayu Aim-X, claims that it is a fully 'made in India' UAV platform built specifically for the Indian scenario and can be used to launch a cross-border surgical strike. Usmani's Mumbai-based start-up, Jatayu, has been working on heavy-lift drones. She adds that the Jatayu Aim-X can be used for surgical strikes, anti-poaching, anti-piracy and burst fire operations. "We hope our UAV platform will help save lives at the front. We have done field trials for the army with 40-42 kilos of payload," Alina says. The drone has an independent ground control station and can be controlled remotely via satellite communication. It has 50 minutes of endurance (ability to perform at the optimum level) in high-altitude,